



Self-supervised cardiac ultrasound segmentation

A paper talk on label-free echocardiography chamber masks

Research Presentation Program

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The model object is a weak-label SSL segmentation pipeline

The source pipeline is manual-label-free for training. human segmentations in later figures are visual aids or evaluation references.



The image shows a small, rotated snippet of a table from a document. The table has multiple columns with headers and rows of data. The text is too small and rotated to be legible, but it appears to be a technical or scientific table.

Figure 1 page, Ferreira et al. 2025, CC BY.

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A2C bootstraps the later chamber pipeline

Article

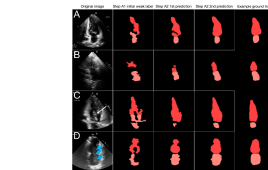
<https://doi.org/10.1038/s41467-025-60405-9>

Fig. 2 | Examples of segmentation and measurement improvement through successive steps of the A2C pipeline for the apical 2-chamber view. Visualization of segmentation for several examples for A2C (A–D). A shows a typical good quality A2C image. The initial weak label (red) requires the bootstrap process but are corrected through successive steps of the pipeline to final prediction (step D). The pipeline performs well even with left atrial enlargement (a noticeable dilated image (B), an image where LV contour (white arrow) obscures the LV lumen (C), and an image with a large LA mass (blue arrowhead) protrudes into the LV (D)). For each of (A–D), complete normal segmentation is shown as a visual aid for the reader. **Left atrial volume (LAV) and stroke volume (SV).** Note that no manual segmentations were used in training the pipeline. Note that final clinical measurements were not made from a single image frame but rather of image frames in a given cardiac cycle, and often multiple views at once. A2C, apical 2-chamber; LV, left ventricle.

To demonstrate incremental impact of each step of the pipeline on predictive performance and generalizability, segmentations from intermediate steps of the pipeline were compared to clinical area measurements on images from the validation dataset (Fig. 3). The r^2 on chamber areas ranged from 0.06 to 0.22 when using initial weak labels, compared to 0.33–0.81 using the full pipeline. Bias and LOA similarly improved with successive training steps.

Pipeline-derived structure and function measurements are comparable to clinical echocardiogram measurements

Pipeline-derived measurements in the all-comers dataset were compared to clinical echocardiogram measurements (Fig. 6, Supplementary Table S1, Supplementary Fig. S3, S4).

Left ventricle. Pearson correlations (r) between the AI pipeline and clinical echocardiogram measurements for LV end-diastolic volume (LVEDV), LV end-systolic volume (LVESV), and LV ejection fraction (LVEF) were 0.84, 0.93, and 0.81, respectively (Supplementary Table S2). r^2 for these measurements were 0.70, 0.86, and 0.65, respectively (Fig. 6). LVEF achieved through the self-supervised AI pipeline is better than those reported from supervised learning¹⁰ (Figs. 6A and S3, Supplementary Table S2). Bland-Altman bias $\pm$ LOA (two standard deviations) for LVEDV, LVESV, and LVEF were -2.6 ± 15.6 mL, -5.5 ± 25.6 mL, and $-5.3 \pm 14.6\%$, respectively, similar to clinician variability studies^{10,11} and consistent with those achieved from supervised learning¹⁰ (Figs. 6A, S3, Supplementary Table S1, Supplementary Fig. S3). Correlations for LV mass were lower ($r = 0.52$, $r^2 = 0.27$), but are still useful to reported benchmarks (Fig. 6B).

Right ventricle. r^2 were 0.69 and 0.71 for RVEDV and RVESV, respectively, indicating a strong correlation¹² between the AI pipeline and

clinical measurements. r 's were 0.63 and 0.64 and bias $\pm$ LOA was -0.86 ± 5.4 mL, -1.6 ± 5.9 mL (Supplementary Table S1, Figs. 6C, S3, Supplementary Fig. S3).

Aorta. The r^2 for left atrial volume was 0.84, showing a very strong correlation. Bias $\pm$ LOA was -0.13 ± 20 mL, consistent with clinical inter-observer bias and with supervised learning performance reported in the literature¹³ (Fig. 6D). Right atrial volume r^2 was 0.76 and bias $\pm$ LOA between AI and clinical measurements was -2.1 ± 21.9 mL (Fig. 6E).

Normal vs. abnormal. For further clinical context, the above measurements were indexed to body surface area where applicable and binned as normal or abnormal according to reference guidelines^{14–16}. Accuracy and Cohen's kappa values are presented in Fig. 6, Supplementary Table S1 and Supplementary Fig. S4. Accuracy ranged from 0.73 for LV mass to 0.97 for LAV. Kappa values ranged from 0.56 to 0.79 showing a moderate to substantial agreement¹⁷ between AI pipeline and clinical echocardiogram measurements. LV mass and RVESV were the exception, where kappa values were only fair.

Taken together, measurements derived from self-supervised learning performed similarly to clinical inter- and intra-observer variability and to measurements derived from supervised deep learning.

Self-supervised segmentation and clinical echo correlate similarly to CMR gold standard

Compared to all-comers, the CMR subset is younger, more male, and has more LV dysfunction (Table 1). Also, due to the nature of different clinical imaging protocols, CMR studies did not include measurements for all chambers. Furthermore, methods for measurement differ

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Figure A2C pipeline improves left and right chamber labels and the progression from weak labels to final predictions on real ultrasound frames.



Noisy echo and missing labels define the failure boundary

Article

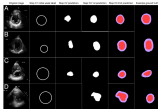
<https://doi.org/10.1038/s41467-020-18405-1>

Fig. 1 Examples of noisy echo and missing labels. **a**, Noisy echo image of the SAX plane for the dataset. **b**, Noisy echo image of the SAX plane for the dataset. **c**, Noisy echo image of the SAX plane for the dataset. **d**, Noisy echo image of the SAX plane for the dataset.

b, Noisy echo image of the SAX plane for the dataset. **c**, Noisy echo image of the SAX plane for the dataset. **d**, Noisy echo image of the SAX plane for the dataset.

Discussion

Segmentation is a global, critical, and challenging task in ultrasound. The use of task-free data learning provides a high-quality solution. The main challenge is to provide a consistent, it is difficult to self-supervised learning, and yet, supervised learning would require even when high-quality ground truth is available. We solve this problem by developing a global self-supervised segmentation of cardiac structures from echocardiogram videos using a novel, global, self-supervised learning framework. We solve this problem by developing a global self-supervised segmentation of cardiac structures from echocardiogram videos using a novel, global, self-supervised learning framework. We solve this problem by developing a global self-supervised segmentation of cardiac structures from echocardiogram videos using a novel, global, self-supervised learning framework.

dataset are missing many segmentations both systematically (e.g., right heart, left atrium and ventricle) or randomly in the absence of labels. In addition, the segmentation is often of poor quality. While supervised learning requires labels, global self-supervised learning does not. Global self-supervised learning is able to segment all structures in an image and learn to be accurate, regardless of the quality of the labels. In the same way, the efficiency and interpretability are greater than other cardiac models^{10,11}. Future research requires deep self-supervised learning. In future research, we will use global self-supervised learning to segment all structures in an image and learn to be accurate, regardless of the quality of the labels. In the same way, the efficiency and interpretability are greater than other cardiac models^{10,11}. Future research requires deep self-supervised learning. In future research, we will use global self-supervised learning to segment all structures in an image and learn to be accurate, regardless of the quality of the labels.

Supplementary Information | <https://doi.org/10.1038/s41467-020-18405-1>

The SAX page includes a technically difficult image and the discussion keeps frame selection and incomplete external labels as limits. Figure 4 page plus discussion limits.

The claim is scalable measurement, not label magic

Failure evidence

Noisy ultrasound boundaries and clinical frame selection can break a simple segmentation-quality story.



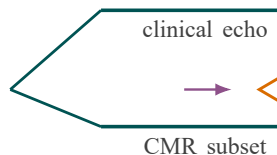
Observed limit

motivates

Manipulate sampling

- preserve view labels
- check chamber anatomy
- compare measurements

Comparator arms



Decision rule

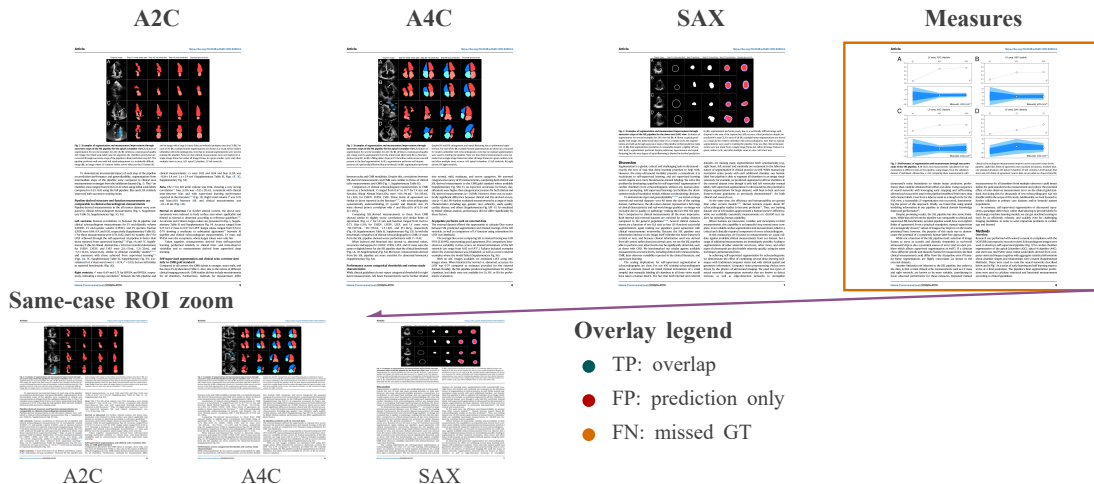
go/no-go

Accept the pipeline only if real echo pixels and clinical measurement semantics remain tied to the article.

The paper-specific conclusion is that label-free training can scale segmentation, while validation remains clinical and anatomy-specific.



A2C, A4C, and SAX must not be conflated



All panels are source article pixels; the semantic risk is mixing view labels or treating clinical measurements as single-frame manual labels.